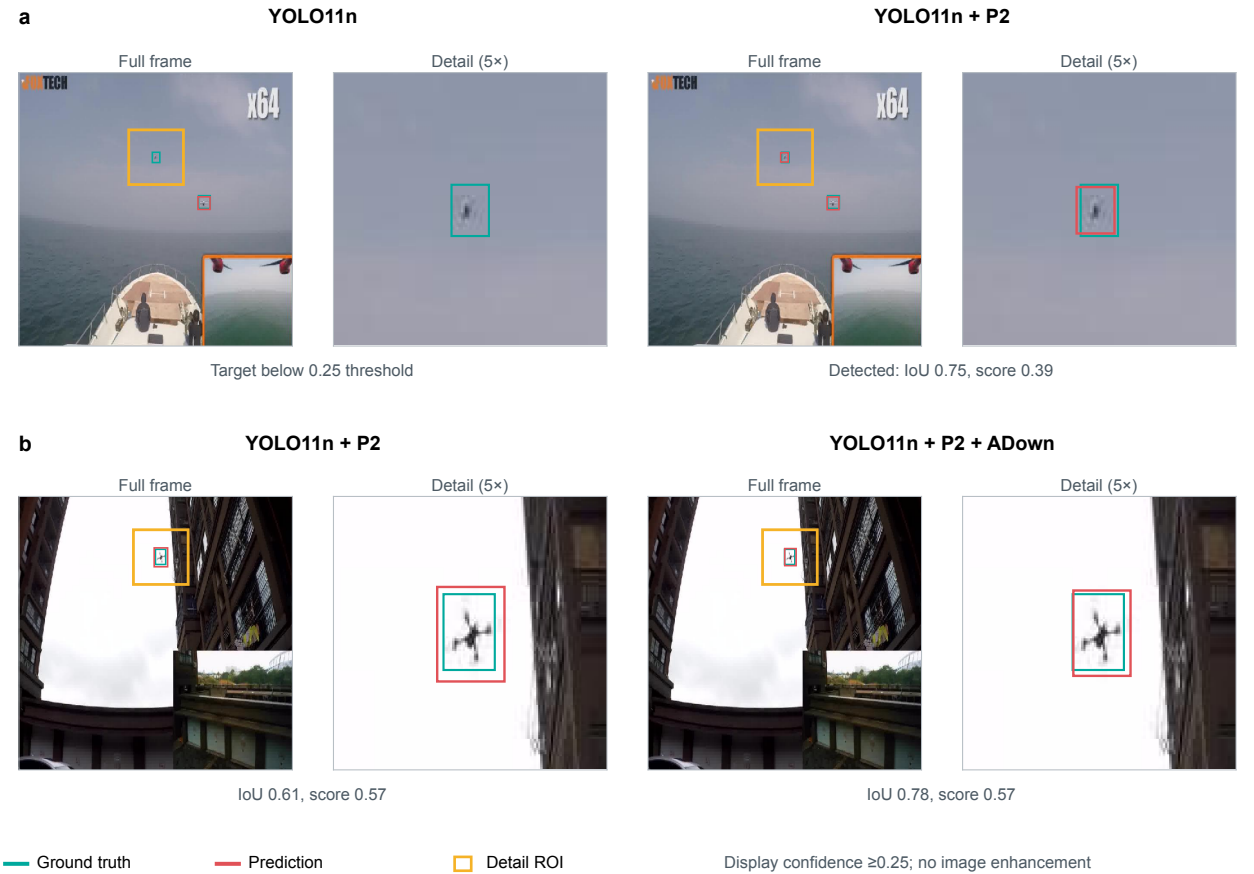

a
YOLO11n
YOLO11n + P2
Full frame
Detail (5×)
Full frame
Detail (5×)
Target below 0.25 threshold
Detected: IoU 0.75, score 0.39
b
YOLO11n + P2
YOLO11n + P2 + ADown
Full frame
Detail (5×)
Full frame
Detail (5×)
IoU 0.61, score 0.57
IoU 0.78, score 0.57
Ground truth
Prediction
Detail ROI
Display confidence ≥0.25; no image enhancement